Robert Burdsall

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Summary

Electrical Engineering student at NC State with hands-on industrial electrical experience: control panel design and fabrication, power distribution hardware, schematic capture, and documentation built for the technicians who service the equipment. Built 5+ control panels and led an automated inspection machine running 120 parts/min from concept through commissioning. Seeking an electrical engineering internship in power systems, electrical distribution, or energy.

Education

North Carolina State University — Raleigh, NC
Bachelor of Science in Electrical Engineering, GPA: 3.5 — *Expected Dec. 2028*

- **Relevant Coursework:** Introduction to Signals, Circuits and Systems; Fundamentals of Logic Design; Computer Systems Programming; Introduction to Computer Systems

North Carolina School of Science and Mathematics — Morganton, NC
High School Diploma, Computer Science Focus, GPA: 4.8 — *Aug. 2023 – May 2025*

Experience

IDEAL Fastener Corporation — Oxford, NC
Electrical Engineering Intern — *May 2026 – Aug. 2026*

- Designed and fabricated 5+ electrical control panels, laying out DIN rail, relays, power supplies, and disconnects and performing full wiring and terminations out to proximity sensors, laser sensors, and solenoid valves.
- Produced TinyCAD schematics for every panel so plant technicians could replicate, service, and troubleshoot the hardware from drawings alone.
- Owned an automated inspection machine end to end — concept, electrical design, build, and commissioning — running 120 parts/min with Keyence machine vision, vacuum ejection, solenoid valve sequencing, a DOPSoft HMI, and PLC quality tracking with good/bad rate alarms.
- Established automated defect detection for the first time on 3 existing production machines, reaching 100% inspection coverage by configuring Keyence IV3/IV4 vision systems and defining defect recognition criteria.
- Programmed Omron PLCs in CX-Programmer, implementing ladder logic control sequences and interlocks.

FIRST Robotics Competition, Team 9150 Labyrinth Robotics — Morganton, NC
Software Developer (2024–25); Lead Programming Mentor (2025–Present) — *Aug. 2024 – Present*

- Designed and implemented the full robot control software in Java using WPILib, integrating vision and dual scoring systems on hardware wired, built, and debugged under a compressed build-to-competition schedule.
- Achieved the 3rd-highest scoring performance across 2025 North Carolina events, including the UNC Asheville District Event.
- Lead and mentor 25+ students, teaching Java, WPILib, wiring, and iterative test and debug of a competition robot.

North Carolina School of Science and Mathematics — Morganton, NC
Computer Science Teaching Assistant — *Aug. 2024 – May 2025*

- Taught 150+ students programming fundamentals in Java and Python.
- Built a cross-platform resource tracking application in Flutter and Dart in collaboration with faculty.

Projects

Aurum | *Dart, Flutter, Python, Express.js, TensorFlow* — Sep. 2024 – Dec. 2024

- Built a cross-platform mobile application for real-time coin appraisal, running on-device inference with TensorFlow Lite.
- Developed the Express.js backend with JSON APIs and JWT authentication for live value retrieval and portfolio tracking.

Organizations

Boy Scouts of America, Troop 222 — Huntersville, NC
Eagle Scout; Senior Patrol Leader and Assistant Scoutmaster — *2015 – Present*

- Mentor 30+ Scouts as Assistant Scoutmaster and lead outdoor programs focused on leadership, teamwork, and problem-solving.

Skills

Electrical Design: Control panel layout and fabrication, wiring and terminations, power distribution hardware (relays, power supplies, DIN rail, disconnects), sensor and actuator wiring, schematic capture (TinyCAD)
Controls and Instrumentation: Omron PLC programming (CX-Programmer), HMI development (DOPSoft), machine vision (Keyence IV3/IV4, IV Smart Navigator)
Design and Fabrication: SolidWorks (Essentials for Part Design certificate), 3D printing (Prusa Slicer, Cura)
Languages: Java, Python, C/C++, Dart, SQL, JavaScript, HTML/CSS
Developer Tools: Git, Docker, VS Code, IntelliJ, TensorFlow, Google Cloud Platform, Android Studio, Xcode